Public Assembly and Mass-Gathering Venues Facility-Class Design Basis

The First Facility Class of the Commercial Facilities Sector Design Basis

DBA-CF-PA-FC1 — Facility-Class Design Basis, DBA-CF Series

Issued Under: Commercial Facilities Sector Framework (DBA-CF_Sector_Framework v0.4); DBA-MA Parent Framework

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose and Scope

1.1 Purpose

This document sets out the design basis for public assembly and mass-gathering venues, the first facility class in the DBA-CF series. The *Commercial Facilities Sector Framework* (DBA-CF_Sector_Framework) states the sector's design basis in general terms, and this document applies it to one kind of facility: the stadium, arena, theater, convention center, theme park or event site at which thousands of people gather in one place at one time. It names the events such a facility must withstand, traces how each event reaches the people in it and the population that may come to depend on it, sets the parameters that size the class's protection, and states what must be true of a venue, of the crowd it admits and of the arrangements by which it is run for the people in it to leave it alive and for the venue to serve when a disaster turns it into a refuge. It describes no particular venue. A design basis for a specific venue or event starts from this document and adds only what is particular to it.

The framework sets two roles apart in this sector, beyond the soft-target framing it accepts as far as it goes. The first is the disaster-support role: the large venue is the nation's mass-care infrastructure, the shelter and reception center when a hurricane, flood or wildfire forces evacuation, so that its loss is the loss of a refuge at the moment it is most needed (framework §1.5). The second is the convergence-node role: the large urban venue often sits where several other sectors stack within a small radius, so that an event there cascades across them (framework §1.5). It therefore requires three consequence treatments, the mass-gathering life-safety consequence, the disaster-support consequence and the convergence-node consequence (framework §3.6). The framework names this class as the one on which both roles concentrate, and the highest-consequence class of the sector (framework §3.2). On the author's decisions of 14 October 2026 the class is the sector's first and proposes three reference events of its own (Section 9.2).

1.2 Scope

The class covers the facilities the framework's register names for the function: sports stadiums and arenas, performing-arts venues and theaters, convention centers, theme parks and large outdoor events (framework §3.2, §8.2). For a facility it covers the structure or site and its means of egress; the crowd it admits, its number, its density and its movement; the spaces in and around the venue where people gather before they are screened and after they leave, whoever owns them; the screening, stewarding and security of the event; the event's plan and the authority within it to stop the show or the flow of the crowd; the medical capacity at the venue; the venue's control and communication systems, from its public address to its access control and ticketing; and, when the venue is used as a shelter, its capacity to hold and sustain the people who come to it. It covers the permanent venue and the temporary event site alike, because a festival built for a weekend in a parking lot or a park holds as many people as a stadium and has none of the stadium's permanent provision.

Three neighbors are separate classes of this series. Hotels and motels, office buildings, multi-unit residential buildings and mixed-use developments are *Lodging and Real Estate* (DBA-CF-LR-FC2). Shopping centers, retail stores, gaming facilities and food service are *Retail and Commerce* (DBA-CF-RT-FC3); the cyberattack on the hotel and casino operations of MGM Resorts in 2023, which the plan for closing the seven sectors named for this sector, belongs to those classes and is held for them (Section 9.2). Private venue and event security and its interface with public safety is the coordination class, DBA-CF-CorpCo. They are covered here only where a public-assembly venue depends on them. Section 8.3 states the boundaries.

Three boundaries with other series are drawn in the framework and taken here as given. The disaster-support role is a shared interface with Emergency Services: the facility and its life-safety systems are this sector's, and the mass-care mission it hosts is governed by the emergency-services and Emergency Support Function 6 framework (framework §6.3). The convergence-node role is a shared interface with every co-located sector, at which this design basis hands off to the cross-sector interdependency analysis (framework §6.3). And the planned gathering is a shared interface with Emergency Services drawn by who acts, on the author's decision of 7 October 2026: the venue's and the promoter's side belongs to this class, and the framework says that its full statement is owed by this class (framework §6.3, §9). Section 4.5 states it.

What makes a facility a member of this class is the same at every venue: it gathers more people in one place than the ordinary arrangements of a building can protect, it is open to them by design, and it may be called on to shelter a population when the population has nowhere else to go. The particular venue, event and crowd differ from site to site, and that detail belongs in each facility's own design basis.

1.3 The Reference Facilities

Three kinds of facility are used as reference cases. Each was chosen because it shows a different way the people in a venue, or the population that depends on it, can be lost.

The enclosed arena with shared public spaces holds an audience inside a screened perimeter, and empties it through spaces it does not own alone: a foyer, a concourse, a link to a railway station, shared with other owners and with the public. It is the governing configuration for the deliberate attack and for the edge of the perimeter. The bombing at the Manchester Arena on 22 May 2017, in a foyer outside the screened perimeter as the audience left a concert, conferred by the framework as DBA-CF-RE-1 (framework §9.1), is the case (Section 5.3).

The temporary event site is a festival or concert built for a few days on open ground, with its stages, barriers, entrances and medical posts brought in and taken away, and a crowd as large as a stadium's held by arrangements that exist only for the event. It is the governing configuration for the crowd itself, for its density and for the authority to stop the show. The crowd crush at the Astroworld festival in Houston on 5 November 2021, conferred by the framework as DBA-CF-RE-2 (framework §9.1), is the case (Section 5.2).

The stadium as refuge is a large venue opened as a shelter when a disaster forces a population from its homes. It is the governing configuration for the disaster-support role of the framework, and for the venue whose design basis is set by a disaster other than its own. The use of the Louisiana Superdome as the refuge of last resort in New Orleans during Hurricane Katrina in August and September 2005, conferred by the framework as DBA-CF-RE-3 (framework §9.1), is the case (Section 5.5).

1.4 Class Type

This is a physical facility class under the author's determination of 30 September 2026 (Facility-Class Types Are Physical, Coordination and Logical). The harm it governs runs through structures, sites and crowds, and through the egress, protection and sustainment the venue provides. In the framework's structure it sits under Public Assembly and Mass-Gathering Venues (PA), the first of the sector's three function classes (framework §3.1, §3.2). It carries one thing that sets it apart from the physical classes before it: its hazard is the crowd. A crowd at a dense enough packing can kill the people in it with no fire, no structural failure and no attacker, and the class therefore states the crowd's density, and the authority to relieve it, as design-basis quantities alongside the venue's structure and its protection.

1.5 Relationship to Governing Documents

This instrument is issued under the Commercial Facilities Sector Framework and inherits the DBA-MA parent framework. It does not repeat the framework's taxonomy, bounding principles, three consequence treatments or Bright Line and Command Broker architecture; it inherits them, and Section 9 is the required inheritance map. Where the framework is still a skeleton (framework §4, §5 and §7), this instrument develops the class-level content ahead of it and records in Section 9 what the framework should take up. Section references of the form framework §N resolve against the Commercial Facilities Sector Framework in force.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application and the obligation it creates — is defined in FD-BL (*The Bright Line*), which governs. The Broker role that discharges that obligation is defined in FD-BR (*The Broker*). Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD (*Layer Decomposition and Determinism*). Local sovereignty is defined in FD-LS (*Local Sovereignty*). Two determinations bind under FD-BL: FD-BL-D1 (*Mode of Discharge*) governs how the authorization obligation may be discharged, and FD-BL-D2 (*Indication Integrity*) governs the integrity of the basis on which an authorization is granted. FD-EV (*Envelope*) reserves the term envelope. Placement is on the consequence of the individual action, assessed per action; this instrument applies that definition and does not vary it. The Foundational Definitions are cited without a version token, and FD-REG (*Governing Stack Register*) is authoritative for their version state.

2. The Public Assembly and Mass-Gathering Venue Facility Class

2.1 Defining Characteristics

Six characteristics define the class, and each is a reason the loss of control of a venue reaches farther than the loss of most buildings.

The first is the number. A venue gathers tens of thousands of people in one place at one time, by design, and every hazard the venue faces is multiplied by that number. The framework names the mass-gathering life-safety consequence as the sector's most visible concern (framework §1.5).

The second is the crowd as a hazard of its own. Above a certain density a crowd stops behaving as people moving and begins to behave as a mass, in which pressure is transmitted through bodies and people can die standing up, without any other cause. A venue's design basis is not met by its structure and its fire protection alone; it must also hold the density of its crowd below the point at which the crowd itself becomes the hazard.

The third is the edge. A venue screens the people who enter it, but its crowd gathers before the screen and disperses after it, in spaces the venue shares with others or does not own: a plaza, a foyer, a station link, a queue on a public street. An attacker who cannot pass the screen need only wait outside it, at the time the crowd is densest and least protected, which is when it leaves.

The fourth is the temporary event. Much of the class is built for a single event and taken down after it, under a permit, by a promoter and contractors who will not run the next event on the same ground. Its egress, its barriers, its medical provision and its command structure exist only for the event and are tested only by it.

The fifth is the refuge. The venue that holds a crowd for an evening can hold a population for a week, and it is called on to do so when a disaster has taken everything else. Its design basis in that role is set by a disaster that is not its own, and its value is greatest when the systems on which it would rely, power, water and transport, have failed around it (framework §1.5).

The sixth is the node. A large urban venue is often built on top of, or beside, the transport, power and communications of a city, so that an event at the venue reaches them and an event in them reaches the venue (framework §1.5). The venue's crowd is, at the end of an event, the largest single load the transport node beneath it carries.

2.2 The Occupancy and Convergence Sub-Index

The framework sub-indexes the sector's classes by occupancy and by convergence exposure, with the disaster-support and convergence-node dimensions carried as overlays (framework §3.1, §8.2). This instrument states three tiers by occupancy and carries the two overlays separately, because the framework measures them differently (framework §3.6). A facility's tier and its overlays are stated separately and are not averaged.

By occupancy, the first tier is the venue or event that admits more people at once than the number the jurisdiction sets for the largest class of assembly, or whose crowd at its densest point is held by temporary arrangements alone; the second tier is every other venue or event whose occupancy requires trained crowd management under the life-safety code; and the third tier is every other facility in the class. The disaster-support overlay is stated for a venue that is designated, or is likely to be used, as a shelter or reception center, by the population it is expected to hold and for how long. The convergence overlay is stated for a venue by the number of other sectors' facilities within the radius the jurisdiction sets, and by whether the venue's crowd discharges through them. A small theater can be third-tier and carry no overlay; a stadium can be first-tier, a designated shelter and a convergence node at once.

2.3 Deployment Typologies and Size Range

The class spans several kinds of facility. A stadium or arena is a permanent structure with fixed seating, a screened perimeter, its own security and medical posts and, often, a control room from which its operator watches its crowd and its systems. A theater or performing-arts venue holds fewer people in a denser space with fewer exits. A convention center holds a changing population across large open halls, and is the venue most often opened as a shelter. A theme park spreads its crowd over a site for a whole day. And a temporary event site, a festival or an outdoor concert, holds a crowd as large as a stadium's on ground that was a parking lot or a park the week before, under a permit, with a promoter and a security contractor who may never have worked together.

The range is wide. At one end is a stadium seating seventy thousand above a commuter rail station in the center of a city; at the other a hall holding a few hundred for a concert. The class is written to fit both, because the consequence of either is measured in the lives of the people in it.

2.4 Consequence Context

The class's consequence runs on the three treatments the framework requires (framework §3.6). For the mass-gathering life-safety consequence it is the harm to the people in the venue, measured against its occupancy and against the capacity of its egress under a deliberate-event condition. For the disaster-support consequence it is the shelter capacity lost, measured against the population that depends on it in the disaster that calls on it. For the convergence-node consequence it is the aggregate consequence across the co-located sectors, which this instrument states and hands to the interdependency analysis.

The record shows the first two. At 22:31 on 22 May 2017 a bomb was detonated in the City Room, a foyer outside the screened perimeter of the Manchester Arena through which the audience of a concert was leaving, killing twenty-two people, among them children, and injuring hundreds more (Section 5.3). On 5 November 2021, at a festival attended by about fifty thousand people in Houston, a crowd pressed toward a stage, and ten people, aged nine to twenty-seven, died of compression asphyxia, as the medical examiner found (Section 5.2). And in August and September 2005 more than twenty thousand people sheltered in the Louisiana Superdome after

Hurricane Katrina, which tore the covering from its roof and cut its power, so that they lived for nearly a week without working plumbing, as the Senate's investigating committee found (Section 5.5).

3. The Facility-Class Design Envelope

3.1 Why Public Assembly Is the First Class

Three reasons put this class first. It is the function the framework names first, and it is the class on which both of the framework's overlays concentrate, making it the highest-consequence and most analytically distinctive class of the sector (framework §3.2). It is the class whose side of the planned-gathering interface with Emergency Services the framework already owes, so that the interface the Emergency Services series stated from its side in October 2026 cannot be closed until this class is written (framework §6.3, §9). And it is the class whose events the plan for closing the seven sectors named: a bombing at an arena and a crowd crush at a festival, together, on the author's decision of 14 October 2026, with a stadium's failure as a refuge. The cyberattack on MGM Resorts, which the plan also named, struck hotel and casino operations and belongs to the lodging and retail classes, and is held for them (Section 9.2).

3.2 The Three Consequence Chains

Every event in the class reaches people by one of three chains, which are the framework's three consequence treatments stated as chains.

Chain L, the life-safety chain, begins with something that harms the people in a venue or at its edge: an attack, a fire, a structural failure, or the crowd itself reaching a density at which it crushes the people in it. Its consequence is measured in lives and injuries, and it is bounded by the venue's occupancy and density, by the reach of its perimeter, by the capacity of its egress, by the authority within the event to stop what is happening, and by the medical capacity at hand. It is the framework's mass-gathering life-safety consequence (framework §3.6).

Chain R, the refuge chain, begins with the loss or failure of a venue in its disaster-support role: the venue is damaged by the disaster it shelters people from, loses the power, water and sanitation it needs to sustain them, cannot be supplied, or cannot be evacuated when it must be. The people in it suffer, and the population that would have come to it has nowhere to go. Its consequence is measured in the shelter capacity lost against the population that depends on it, and in the time the people in it are held in conditions the venue cannot sustain. It is the framework's disaster-support consequence (framework §3.6).

Chain N, the node chain, begins with an event at a venue that is a convergence node, or at the co-located infrastructure, and propagates between them: an attack on the venue that closes the station beneath it, a failure of the station that traps the venue's crowd, a loss of power that takes both. Its consequence is the aggregate across the co-located sectors, and it is not a single-facility

measure at all (framework §3.6). This instrument states the chain and its parameter, and hands its assessment to the cross-sector interdependency analysis (framework §6.3).

The three chains are joined by the crowd. The crowd at the end of an event is densest at the edge of the venue and at the node beneath it, and the population in a refuge is a crowd that cannot go home. And they are joined by timing: an attack timed to the end of an event, or to the hours when a venue is full of the displaced, opens more than one at once.

3.3 Design Parameters at Class Resolution

Six parameters govern the envelope.

The first is the occupancy and egress time: the greatest number of people the venue or event admits at once, and the time to clear each part of it, and the whole, through the egress available under the design-basis event, including egress lost to the event itself. It is this instrument's statement of the framework's measure for the life-safety consequence, occupancy and egress capability under a deliberate-event condition (framework §3.6).

The second is the crowd density: the greatest number of people per unit area the event's plan allows at any point where the crowd can gather, and the means by which it is measured and held below that number during the event. The crowd-safety literature places the onset of dangerous crowd pressure at densities of about five people per square meter and above, and the class does not take a value from it; it requires that each event state its own limit and how it is held.

The third is the perimeter reach: the extent of the spaces in which the event's crowd gathers, queues or disperses that lie within the screened perimeter and under the event's security, against the extent that lie outside it, and for each space outside it, who is responsible for its security. It is the parameter on which the record of Section 5.3 turns. The watch on each space outside the perimeter is confirmed when it is needed: at each entry and egress the venue confirms that the assigned watch on each shared space is in place, and reports a gap at once as a condition to act on, because at Manchester four British Transport Police officers had been told to be in the City Room at the end of the concert and none was there (*The Bombing at the Manchester Arena* F-1, R-1; noted to CorpCo).

The fourth is the stop time: the time from the moment anyone in the event's structure recognizes that people are in danger to the moment the show is stopped, the crowd's flow is halted or reversed, and the people in danger are reached, including the time to tell the person who can stop it and the time for that person to decide. It is the parameter on which the record of Section 5.2 turns. Beside it the class states a report time, the time from a concern raised at the edge to a person with authority to act, across every organization at the event, because at Manchester a member of the public's concern reached stewards and went no further (*The Bombing at the Manchester Arena* F-2, R-2; noted to CorpCo). The stop time is measured from the first of the stated triggers of Section 4.4.

The fifth is the shelter capacity and sustainment: for a venue with the disaster-support overlay, the population it can hold, the number of days it can sustain them with power, water, sanitation, food and medical care independent of the systems the disaster is likely to take, the disasters it can withstand while doing so, and the time to evacuate it when it can no longer. It is this instrument's statement of the framework's measure for the disaster-support consequence (framework §3.6). With the capacity the design basis states the admission stop, who decides it, where those turned away go and what that place provides, because the Superdome was overcrowded by the Tuesday and about nineteen thousand people gathered at the Convention Center, never planned as a shelter, without supplies until the Friday (*The Louisiana Superdome as Refuge of Last Resort in Hurricane Katrina* F-1, R-1; noted to *Emergency Management and Coordination*). And the means of evacuation are credited only if committed before the refuge opens, with the dispatch trigger and authority stated and the time from trigger to an empty refuge (*The Louisiana Superdome as Refuge of Last Resort in Hurricane Katrina* F-2, R-2; noted to *Emergency Management and Coordination* and the Transportation series).

The sixth is the convergence count: for a venue with the convergence overlay, the other sectors' facilities within the radius the jurisdiction sets, and the share of the venue's crowd that discharges through each. It sets the scope of Chain N, and the framework hands its assessment to the interdependency analysis.

This instrument states the parameters and their governing logic. Their quantitative values for a venue or event await validation against what the venues, promoters and jurisdictions already hold: the occupancy and egress calculations made under the building and fire codes, the event plans and permits, the vulnerability assessments CISA offers to venue owners, and, for shelters, the jurisdiction's mass-care plans. The figures quoted are their sources' own.

4. The Crowd, the Perimeter and the Refuge

4.1 Status and Use

Every event in this class reaches people through one of three things the venue keeps: the crowd, its number, density and movement and the authority to act on it; the perimeter, the reach of the event's protection to every space where its crowd gathers; and the refuge, the venue's capacity to hold and sustain a population when a disaster calls on it. They are the class's governing pathway, and this section states what the class requires of each. It follows the pattern of the seat, the act and the records in *Government Operations and Continuity* (DBA-GF-GO-FC1) and of the receiving function in *Care Delivery* (DBA-HPH-CD-FC1), because the crowd, the perimeter and the refuge are the venue's interfaces with the people it gathers and with the attacker or disaster that would harm them.

4.2 The Crowd as a Design-Basis Quantity

A venue's occupancy is usually regarded as a matter of the fire code, set by the area and the exits. In this class the crowd is a design-basis quantity, because the crowd can be the hazard. The

class requires that a first- or second-tier event's design basis state the occupancy and egress time of Section 3.3, the crowd density limit at every point where the crowd can gather, above all at the front of a stage and at the points where it enters and leaves, and the means by which density is watched and held below that limit during the event: the layout of barriers and pens, the number of people admitted to each area, and the people and cameras that watch them. It requires that the limit hold for the crowd's behavior as the event's own nature makes it likely, so that an event whose audience is expected to surge toward the stage is planned for the surge. The record of Section 5.2 shows a crowd that reached a fatal density in front of a stage at an event for which no occupancy load had been issued and whose plan did not hold it below one.

4.3 The Perimeter Reaches the Crowd

A venue's screening is usually placed at its doors. In this class the perimeter is set by the crowd and not by the building, because the attacker can choose where the crowd is. The class requires that a first- or second-tier venue's design basis state the perimeter reach of Section 3.3: every space in which its crowd gathers, queues or disperses, and for each space outside the screened perimeter, who is responsible for its security, what watches it, and how the venue, the other owners of the space and the police who patrol it act together there. It requires that a shared space have a stated security arrangement, agreed among those who own it, use it and patrol it, before the event, and that the times when the crowd is densest at the edge, at entry and at egress, be the times when the edge is most closely watched. The record of Section 5.3 shows an attack in a foyer that lay outside the screened perimeter, that was the responsibility of several organizations whose arrangements were not coordinated, and that was not watched at the moment the attack was made.

4.4 The Authority to Stop

An event is run by a promoter, a venue operator, a security contractor, a production team and the performers, and the people best placed to see danger in the crowd are seldom the people who can stop the show. The class requires that a first- or second-tier event's plan state who may stop the show, halt the crowd's flow or open the barriers, on what judgment and on whose word; that the authority to stop rest with a person who can see the crowd or is told at once by those who can; that no commercial or production consideration be a reason to continue once that person judges that people are in danger; and that the stop time of Section 3.3 that results be stated and exercised. The record of Section 5.2 shows a show that continued, on the police department's account, after those running it had been told that people in the crowd were dying. The stop has triggers: the plan and the permit state triggers that stop the show without further judgment, among them people down and any declared mass-casualty incident, because at Astroworld, on the police department's report as reported, warnings reached the event's structure from 9:00 p.m., the camera operator's radio call came at about 9:25 p.m., the fire department declared a mass-casualty incident at 9:38 p.m. and the show ended at about 10:10 p.m. (*The Crowd Crush at the Astroworld Festival* F-1, R-1; noted to CorpCo and *Response Forces and Stations*). The

Texas task force placed stop authority with a production representative in a unified command, on triggers agreed in advance in the permit.

4.5 The Planned Gathering: the Venue's Side of the Interface

The framework draws the line at a planned gathering by who acts, on the author's decision of 7 October 2026, and says that this class owes the full statement of the venue's and the promoter's side (framework §6.3, §9). Emergency Services states the response's side: a plan joining every discipline that would answer a mass-casualty incident at the gathering, a staffed dispatch position for the event, and a check-in point on every approach (*Response Forces and Stations* §4.7, §8.3). This class states the venue's.

Above the occupancy the jurisdiction sets, the class requires three things of the venue and the promoter. First, medical capacity and supplies at the venue sized for the mass-casualty surge the event could produce, which for an event of tens of thousands is not the surge of an ordinary event day: enough trained people, enough supplies to treat bleeding and crush injury, and the means to move casualties out of the crowd to the point where the response can take them. Second, the duty to bring every responding discipline, law enforcement, fire and rescue and emergency medical services, into the event's plan before it opens, so that the response knows the event is taking place, where its entrances, medical posts and command post are, and how it will be told of a mass-casualty incident. Third, a single point within the event's own structure, joined to the response's command, at which the event's security, medical and production leads are present and from which the stop of Section 4.4 can be ordered. The shooting at the Route 91 Harvest festival in Las Vegas in October 2017 is read for the planning of a festival with the fire service by the Emergency Services series from its side (*Response Forces and Stations* §4.7); the record of Section 5.2 shows an event at which those who could stop it were not those who saw the danger. The plan binds every authority that permits or answers: it names the permitting authority and every responding authority, and where they differ each agrees the plan before the event opens, because Astroworld was permitted by the county while the city's 911 answered it and the county's fire marshal disputed the county's jurisdiction (*The Crowd Crush at the Astroworld Festival* F-2, R-2; noted to *Response Forces and Stations* and *Emergency Management and Coordination*).

4.6 The Venue as Refuge

A venue opened as a shelter is usually regarded as a building lent to the mass-care mission. In this class the refuge is a design-basis role, because the disaster that calls on it is also the disaster that threatens it. The class requires that a venue carrying the disaster-support overlay have a design basis that states the shelter capacity and sustainment of Section 3.3: the disasters it is expected to shelter people from, and whether its structure, its roof and its site withstand them; the power, water, sanitation, food and medical care it can provide, for how many people, for how many days, independent of the systems those disasters are likely to take; and the time and means to evacuate it when it can no longer sustain them. It requires that a venue not be designated as a

refuge for a hazard it cannot withstand, and that, where no better refuge exists and the jurisdiction designates it as a refuge of last resort, the shortfall be stated so that the mass-care plan does not count on what the venue cannot give. The mass-care mission is Emergency Services', and the facility's capacity to host it is this class's (framework §6.3). The record of Section 5.5 shows a refuge of last resort in a floodplain, whose roof was torn and whose power failed, holding more than twenty thousand people for nearly a week. The design basis states where the overflow goes and credits evacuation only from means committed before the refuge opens (Section 3.3).

5. Design Basis Events: The Envelope

The events below are the class's design-basis envelope, each stated with the event postulated, the consequence threshold, the function the class requires and the duration for which it is required. Each is marked by the chain or chains it opens. Sections 5.1 and 5.2 are non-adversarial; Sections 5.5 and 5.6 may be either; Sections 5.3, 5.4, 5.7 and 5.8 are adversarial. Inclusion is by life-safety, disaster-support or convergence consequence severity, not likelihood (framework §4.1).

5.1 Fire or Structural Failure at a Full Venue

Postulated event a fire, the collapse of a structure, a stand, a stage or a roof, or the failure of a temporary structure under wind, occurs while a first- or second-tier venue or event is full.

Consequence threshold harm to the people reached by the fire, smoke or collapse, and to those caught in the crowd's flight from it (Chain L).

Required function the egress available after the event clears the venue within its stated egress time; the fire protection and the release of the means of egress act independently of any network an attacker could reach (Section 7.2); and temporary structures are designed and inspected for the loads, wind among them, the event's site and season can bring.

Design-basis duration until the venue is cleared and the injured are reached.

5.2 Crowd Crush

Postulated event the density of the crowd at some point in a first- or second-tier venue or event rises, through surge toward a stage, a blocked or narrowed exit, a sudden movement or an entry that admits more people than an area can hold, to the point at which people in it are crushed or cannot breathe.

Consequence threshold deaths and injuries by compression in the crowd, with no other cause (Chain L), and, where the crowd's discharge loads a convergence node, the closure of the node (Chain N).

Required function the density limit is stated and held, and density is watched at every point where it can rise (Section 4.2); the authority to stop the show and to halt or reverse the crowd's

flow rests with a person who sees the crowd or is told at once, and the stop time is stated and exercised (Section 4.4); and medical capacity at the venue can reach people within the crowd (Section 4.5).

Design-basis duration from the first sign of danger until the crowd is relieved and the injured are reached.

The case is the crowd crush at the Astroworld festival at NRG Park in Houston on 5 November 2021, conferred by the framework as DBA-CF-RE-2 (framework §9.1), whose instrument records it as non-adversarial, following under the RECORD PRIMACY convention of the DBA Reference Event Register the grand jury's finding that no crime occurred. The festival was held in a parking lot at the park, and about fifty thousand people attended, as reported. During the headlining performance the crowd pressed toward the stage, and ten people, aged nine to twenty-seven, died; the medical examiner found that each died of compression asphyxia. Hundreds more were injured. The Texas Governor's Task Force on Concert Safety, in its report of April 2022, found that no occupancy load had been issued for the event, which is typically determined by the fire department, that the permitting process is inconsistent across the state, and that events staged in a parking lot such as this one require unique contingency plans, and recommended a uniform permitting template and a unified command and control group on site, with authority to pause or stop a show. The Houston Police Department's report of its investigation, released in July 2023, records, as reported, that a worker on the event's camera crew radioed the production trailer during the performance that people were dying, and that the show continued for some time after; a grand jury in Harris County had declined in June 2023 to indict anyone investigated, and the District Attorney stated that the grand jury had found that no crime occurred and that no single individual was criminally responsible; the police investigators attributed the deaths to compaction of the crowd, as reported. Civil claims followed. No individual is named here, and the performers and organizers are referred to by role.

The design-basis reading is that a crowd reached a fatal density in front of a stage at an event built for a weekend on open ground, that the arrangements which might have held the crowd below that density, a permit kept, a density watched, barriers that let people out, did not, and that the information that people were dying reached the event's structure without reaching, in time, a person who could and would stop the show. The class therefore states the density and the stop time as design-basis quantities and places the authority to stop with a person who can see the crowd (Sections 4.2, 4.4). The instrument states the task force's findings, the police department's record and the District Attorney's statement of the grand jury's finding as their authors' own.

5.3 Explosive Attack at the Edge of the Perimeter

Postulated event an attacker carrying an explosive device waits in a space where the venue's crowd gathers outside the screened perimeter, and detonates it when the crowd is densest, at entry or at egress.

Consequence threshold mass casualties among the crowd at the edge, many of them in a space the venue does not own alone (Chain L); and, at a convergence node, the closure of the co-located transport and the trapping of the crowd still inside (Chain N).

Required function the perimeter reach is stated, every shared space has an agreed security arrangement among those who own, use and patrol it, and the edge is most closely watched when the crowd is densest there (Section 4.3); a report of suspicious behavior at the edge reaches a person who will act on it; and the medical capacity and the response's plan of Section 4.5 reach the edge as well as the bowl.

Design-basis duration from the opening of the venue to the public until the last of its crowd has dispersed from the edge.

The case is the bombing at the Manchester Arena on 22 May 2017, conferred by the framework as DBA-CF-RE-1 (framework §9.1), whose instrument records it as a non-military adversarial event, with Transportation an impacted sector for the station link. At 22:31, as the audience of a concert was leaving, an attacker detonated a bomb in the City Room, a foyer linking the arena to Manchester Victoria railway station, which lay outside the arena's screened perimeter. Twenty-two people were killed, among them children, and hundreds more were injured. The attacker died in the explosion. A public inquiry under a retired High Court judge reported in three volumes, on security arrangements in June 2021, on the emergency response in November 2022 and on whether the attack could have been prevented in March 2023. The first volume found that the venue operator, SMG, its crowd-management contractor, Showsec, and the British Transport Police each bore responsibility for the security of the City Room, and that their arrangements were not sufficiently coordinated; that the attacker had carried out hostile reconnaissance on at least three occasions and had waited in the City Room, partly in a place its cameras did not see, and should have been identified as a threat on the night; that a member of the public's concern about him had not been adequately acted on; that no police officer was in the City Room when the bomb was detonated, although officers had been told to be there at the end of the concert; and that, while the attacker would probably still have detonated his device had he been confronted, the loss of life and injury would highly likely have been less. It recommended a statutory duty on those responsible for premises to prepare for and protect against terrorist attack. Parliament enacted it as the Terrorism (Protection of Premises) Act 2025, which received Royal Assent on 3 April 2025. The attacker and those found to have assisted him are not named here, and the officials and staff whose conduct the inquiry examined are referred to by role.

The design-basis reading is that the attack was made not inside the screened perimeter but at its edge, in a shared space whose security belonged to several organizations at once and was coordinated by none of them, at the moment the crowd was densest and least protected, and that the inquiry's own finding was that the harm would highly likely have been less had that space been watched. The class therefore sets the perimeter by the crowd rather than by the building and requires a stated arrangement for every shared space (Section 4.3). The instrument states the inquiry's findings as the inquiry's own. The inquiry's findings on the emergency response, in its

second volume, belong with the response's side and are for the Emergency Services series to read.

5.4 Armed Attack or Vehicle Attack on a Crowd

Postulated event an attacker fires on a crowd at a venue or event, from within it or from a position overlooking it, or drives a vehicle into a crowd at the edge.

Consequence threshold mass casualties in the crowd and in its flight (Chain L), and the dispersal of the injured beyond the venue, which loads hospitals and generates false reports (Chain N).

Required function the perimeter reach includes positions overlooking the crowd and the approaches a vehicle could use, with vehicle barriers where the crowd gathers at the edge; the venue's egress lets a crowd flee in more than one direction; and the medical capacity and the plan of Section 4.5 are in place.

Design-basis duration from the opening of the venue until its crowd has dispersed.

The shooting at the Route 91 Harvest festival in Las Vegas on 1 October 2017 is a reference event of the Emergency Services series (DBA-ESS-RE-11), read there for the response, and this class cites it there rather than restating it. Its venue's side, the planning of the festival without every responding discipline, is taken up at Section 4.5.

5.5 Loss of the Refuge

Postulated event a venue opened as a shelter or refuge is damaged by the disaster it shelters people from, or loses the power, water and sanitation it needs to sustain them, or cannot be supplied or evacuated in the time it can hold them.

Consequence threshold harm to the people sheltered, held in conditions the venue cannot sustain (Chain L and Chain R); and the loss of shelter capacity for the population that would have come to it (Chain R).

Required function the venue's shelter capacity and sustainment are stated, it is not designated for a hazard it cannot withstand, and any shortfall in a refuge of last resort is stated for the mass-care plan (Section 4.6); and the time and means to evacuate it are stated and do not depend on the transport the same disaster is likely to take.

Design-basis duration from the opening of the refuge until the last person sheltered is evacuated or goes home.

The case is the use of the Louisiana Superdome as the refuge of last resort in New Orleans during and after Hurricane Katrina, which made landfall on 29 August 2005, conferred by the framework as DBA-CF-RE-3 (framework §9.1), whose instrument records it as non-adversarial, with Emergency Services and Transportation impacted sectors. The House of Representatives' select bipartisan committee, in its report *A Failure of Initiative* of February 2006, found that

Louisiana's plans called for shelters of last resort to be outside the floodplain and wind-resistant, and that the Superdome was in the floodplain and its roof was extensively damaged by wind. The Senate's Committee on Homeland Security and Governmental Affairs, in its report *Hurricane Katrina: A Nation Still Unprepared* of May 2006, found that the storm tore the covering from the roof, caused leaking and cut the power, leaving the plumbing, air conditioning and public address system useless; that more than twenty thousand people, on its figure, stayed in the Superdome without working plumbing for nearly a week, the portable toilets requested for it never having arrived; that five hundred buses assured on the Monday to arrive within hours did not begin to arrive until the Wednesday evening, nor in significant numbers until the Thursday; and, of New Orleans as a whole, that tens of thousands of people waited in what it called unspeakably horrible conditions until as late as the Saturday to be evacuated. The Superdome's last evacuees left on Saturday 3 September, on contemporary reporting. The figures for the number sheltered differ by source: the Senate committee's more than twenty thousand, the House committee's account as *The Failure of Coordination in Hurricane Katrina* (DBA-ESS-RE-8) reads it, more than ten thousand, and an officer of the Guard's reported twenty thousand on the Friday. The committees' findings are stated as their own, and the officials whose decisions they examined are referred to by role.

The design-basis reading is that the region's largest venue was its refuge of last resort for a hazard it could not withstand, that the disaster that filled it also took its power, its water and the transport by which it would be emptied, and that the people in it were held for days in conditions it could not sustain. The class therefore states shelter capacity and sustainment as a design-basis quantity, independent of the systems the disaster takes, and requires that a refuge's shortfall be stated so that the mass-care plan does not count on it (Section 4.6). The disaster-support consequence is the framework's distinctive contribution to the sector, and this is its case.

5.6 Loss of the Node

Postulated event the transport, power or communications infrastructure co-located with a first-tier venue fails, or is attacked, while the venue is full or emptying, or an event at the venue closes the infrastructure.

Consequence threshold the venue's crowd trapped or discharged into an infrastructure that cannot carry it (Chain L), and the aggregate consequence across the co-located sectors (Chain N).

Required function the convergence count is stated, the share of the crowd discharging through each co-located facility is known, and the venue's egress plan does not depend on a co-located facility being open; the assessment of the aggregate consequence is handed to the interdependency analysis (framework §6.3).

Design-basis duration until the venue's crowd has dispersed and the co-located infrastructure is restored.

5.7 Attack on the Venue's Systems

Postulated event an attacker reaches the venue's control and communication systems, its public address and displays, access control, ticketing, lighting, building management or crowd-monitoring cameras, and disables them or uses them to give false instructions, while the venue is full.

Consequence threshold a crowd that cannot be told what to do, or is told the wrong thing, or whose density can no longer be watched (Chain L); and, at a refuge, the loss of the systems that sustain its population (Chain R).

Required function the venue can instruct its crowd and watch its density by means that do not depend on the systems an attacker could reach; the release of the means of egress does not depend on them (Section 7.2); and an instruction to the crowd to move is placed on the consequence of the crowd moving, or not moving, on it, and where that consequence meets the threshold it requires Command Broker authorization (Section 7.3).

Design-basis duration until the venue is cleared or its systems are restored and their integrity shown.

The cyberattack on MGM Resorts in September 2023 struck hotel and casino operations and is held for the lodging and retail classes (Section 9.2).

5.8 Airborne Threat over a Gathering

Postulated event an uncrewed aircraft is flown over or into a crowd at an outdoor venue or event, to deliver a hazard, to cause panic, or to reconnoiter.

Consequence threshold harm to the people struck and to those caught in a panicked flight (Chain L).

Required function the venue's plan states how an aircraft over the crowd is detected and reported, who may act against it and under what authority, and how the crowd is managed if one appears; and an instruction to the crowd prompted by a detection is placed on the consequence of the crowd acting on it, and where that consequence meets the threshold it requires Command Broker authorization (Section 7.3).

Design-basis duration the time the crowd is gathered.

The authority to act against an uncrewed aircraft is held by a small number of federal agencies and, since the SAFER SKIES Act of December 2025, by state and local law enforcement agencies that hold federal certification to exercise it; a venue cannot assume that it can be exercised at its event, and the class requires the plan to state who holds it there. No case is proposed.

Table 5-1. Facility-class design-basis event categories.

Event category	Postulated event (class-specific)	Consequence chain	Framework source
Fire or structural failure at a full venue	Fire, collapse or temporary-structure failure	Chain L	§2.3, §4.1 candidate
Crowd crush	Density rises to the point of compression	Chain L, Chain N	§2.3; developed here
Explosive attack at the edge of the perimeter	Device detonated where the crowd gathers outside the screen	Chain L, Chain N	§4.1 candidate (deliberate mass-casualty attack)
Armed attack or vehicle attack on a crowd	Firing on or driving into a crowd	Chain L, Chain N	§4.1 candidate; DBA-ESS-RE-11
Loss of the refuge	Shelter damaged, unsustained or not evacuated	Chain R, Chain L	§1.5, §4.1 candidate (disaster-support)
Loss of the node	Co-located infrastructure fails or is closed	Chain N, Chain L	§1.5, §4.1 candidate (convergence)
Attack on the venue’s systems	Instruction, access or monitoring systems taken	Chain L, Chain R	§5; developed here
Airborne threat over a gathering	Uncrewed aircraft over or into a crowd	Chain L	developed here

6. Military-Action Extension: DBA-MA for Public Assembly

6.1 MA Extension Applicability Determination

Every first-tier venue, every venue carrying the disaster-support overlay and every venue carrying the convergence overlay performs the facility-level applicability determination the series makes mandatory, under the two-factor test of geolocation and strategic value (framework §7). The framework states a strong sector-level presumption toward applicability for major mass-gathering venues, for facilities in their disaster-support role and for convergence nodes (framework §7). For this class strategic value lies less in the venue than in what it holds: a crowd whose loss would shake a nation, a sheltered population whose loss would compound a disaster, and a node whose loss would cascade. The determination is recorded, it is time-varying and re-triggered on the indicators the DBA-MA parent framework defines, and any scope reduction carries a recorded basis (Section 6.9).

6.2 The Four Characteristics Applied to the Class

Applied to public assembly, the four military-action characteristics take specific form, in the canonical statement of FD-DBL (*Design Basis Layers*), at its §4.4. Intentionality: the adversary chooses the venue whose crowd or sheltered population gives it the effect it seeks. Persistence: the adversary strikes gatherings repeatedly, so that a population stops gathering, and strikes refuges again as they refill. Precision: the adversary chooses the moment when the crowd is densest at the edge, or the shelter is fullest, and the node whose loss traps the most people. Denial: the adversary denies the population its places of gathering and of refuge, so that it has nowhere to assemble and nowhere to go when its homes are lost.

Under the author's rulings of 10 and 13 October 2026, the military-action tier is not met without denial, and all four characteristics must be established on the record; a characteristic not determined counts as not present (*Design Basis Layers* §6.4). None of the three conferred reference events is a military action: the first was a terrorist attack, the second a crowd crush with no attacker, and the third a natural disaster. The tier of each is its instrument's to determine.

6.3 The Refuge Struck, as the Governing MA Scenario

The governing military-action scenario is a strike on a venue in its disaster-support role during a conflict, when it shelters a population displaced by the conflict itself, combined with the denial of the power, water and transport that sustain and would evacuate it. It opens Chains R and L together, and Chain N where the refuge is a node. No credit is taken for a venue's civilian character, for markings that declare it a shelter, or for the protection the law of armed conflict gives it.

The extension requires three things. It requires that a venue designated as a refuge in a region within an adversary's reach have places within it where the people sheltered can be protected from blast and fragments, and that its population not exceed what those places hold. It requires that its sustainment of Section 3.3 not depend on systems the same campaign can reach. And it requires that the mass-care plan spread the sheltered population across refuges, so that no single strike can reach most of it, with the admission stop and the overflow of Section 3.3 stated for each.

The empirical record is not among this class's proposed events, and it is noted rather than proposed. On 16 March 2022 the Donetsk Regional Drama Theater in Mariupol, a performing-arts venue being used as the city's largest shelter during the siege, with the Russian word for children marked on the ground outside it, was destroyed. Amnesty International, in an investigation published on 30 June 2022, concluded that it was struck by a Russian airstrike and that the strike was a war crime, and estimated that at least a dozen people were killed and likely many more; an investigation by the Associated Press in May 2022 put the dead at about six hundred; the city council estimated about three hundred; and a report under the Organization for Security and Co-operation in Europe in April 2022 found the strike most likely an egregious violation of international humanitarian law. Russia's Ministry of Defense denied that its aircraft

had struck ground targets in the city that day and said that a Ukrainian unit had blown up the building. The design-basis reading is that a theater used as a refuge, and marked as one, has been destroyed in a war of this decade with many of the people sheltering in it, and that the nation's own design basis for its refuges must postulate that they can be. The instrument states each account as its own and makes no finding of its own on responsibility or on the number killed beyond what its sources establish.

6.4 The Gathering Struck

The second scenario is a strike on a full venue, by missile, drone or a sabotage team, timed to a national occasion, so that the harm is both the loss of life and the shock to the nation that a crowd at its own celebration was struck. It opens Chain L and, at a node, Chain N. Against it the class relies on the perimeter, egress and medical provision of Section 4, and on the jurisdiction's decision whether to hold the gathering at all when the threat is high, which is a governed decision on a stated basis and is the coordination class's to support.

6.5 The Compound Scenario

The framework names among its candidate bounding events the convergence-node cascade (framework §4.1). For this class its compound form is an attack on a full venue at a node, timed so that the co-located station is closed or struck at the same moment and the venue's systems are taken, so that the crowd is attacked, cannot be told where to go, and cannot leave by the way it came. Against it the class relies on egress that does not depend on the node (Section 5.6), on means of instructing the crowd that do not depend on systems an attacker could reach (Section 5.7), and on the release of the means of egress as a locally sovereign function (Section 7.2).

6.6 Cross-Sector Interaction

Under military action the class's outward consequence runs in the three directions the framework names: the life-safety consequence of a struck gathering, the economic consequence of a population that stops gathering, and the response-reserve consequence of a refuge lost when it is most needed (framework §6.2). Coming in, the class depends on energy, communications, transportation and water for its operation and, as a refuge, for its population's survival (framework §6.1), and on Emergency Services for the response and the mass-care mission. At a convergence node its inward view is a tangle rather than a set of separable chains (framework §6.1).

6.7 The Adequacy Result: Two Conditions

Because the framework measures the life-safety and disaster-support consequences differently and declines to measure the convergence consequence at the scale of a single facility (framework §3.6), the test of adequacy is two tests, and a facility meets the class design basis only if it meets both, or, where it carries no disaster-support overlay, the first. The convergence consequence is not tested here; it is handed to the interdependency analysis (Section 6.8).

The first is what this document calls the Safe Assembly Condition. When the venue is full in the design-basis event, its crowd is held below the stated density at every point, its perimeter reaches every space where the crowd gathers, the authority to stop acts within the stated stop time, the medical capacity at the venue reaches the injured within the crowd and at its edge, and the crowd clears the venue through the egress the event leaves within the stated egress time. The condition bounds Chain L.

The second is what this document calls the Sustained Refuge Condition. When a venue carrying the disaster-support overlay is opened as a refuge in the disaster that calls on it, it withstands that disaster, it sustains its stated population with power, water, sanitation, food and medical care for the stated number of days independent of the systems the disaster takes, and it can be evacuated within the time it can sustain them. The condition bounds Chain R, and it is met only by a refuge whose sustainment has been shown, not assumed.

The protective functions on which a full venue relies in the moment are those that let its crowd escape and keep fire from spreading: the release of locked doors and gates on the means of egress when the fire alarm sounds or power is lost, emergency lighting and wayfinding, fire detection and suppression, and smoke control. Where they are protective functions in that sense they are to be locally sovereign as FD-LS defines the property (*Local Sovereignty* §3.2): a measurement against a setpoint, colocated with the hazard, which no system an attacker could reach can stop, delay or trigger, and whose settings change only through a governed act. Local Sovereignty governs and is not restated here. It describes what protects the crowd when the venue's control systems have failed; it does not decide whether any action needs human authorization (*Local Sovereignty* §4). It has nothing to say about the density of the crowd, which no physical protection relieves by itself.

6.8 Adequacy and the Declared Residual

In the non-adversarial domain, the Safe Assembly Condition is achievable with the Section 4 requirements: a stated and held density, a perimeter set by the crowd, an exercised authority to stop and medical provision sized for the surge. Against a deliberate adversary, adequacy has limits.

Three residuals are declared, in the output form DBA-MA-DiD (*Defense in Depth*) sets at its §3.5. First, an attacker who detonates a device or opens fire before any protection acts leaves a residual in lives at the point of attack, which the perimeter, egress and medical provision reduce but do not remove; its measure is the crowd within reach of the attack, and the response is the perimeter reach and the watch on the edge. Second, a refuge struck in a conflict, or overwhelmed by a disaster beyond its design, leaves a residual in the population sheltered; its measure is that population, and the response is to spread it across refuges. Third, the convergence consequence at a node is a residual that this class cannot assess alone; it is declared here, by the convergence count, and handed to the interdependency analysis, so that it is not mistaken for a consequence the class has answered. These residuals bound the class adequacy claim.

6.9 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter, such as an occupancy credited from the permit rather than from the count at the gate, a density limit credited without the means to watch it, a perimeter credited at the building's doors when the crowd gathers beyond them, a stop authority credited to a person who cannot see the crowd, medical capacity credited for an ordinary day at an event of tens of thousands, a refuge credited for a hazard its structure cannot withstand, or sustainment credited from systems the disaster will take, must be justified against the class baseline, and the larger the relaxation the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis. The framework does not yet state this principle for the sector, and Sections 9.1 and 10.9 record it for the framework to adopt.

7. The Dual Design Basis, Stated in Full

7.1 The Two Design Bases

The facility carries two design bases at once: a physical design basis, defined in this instrument, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here (framework §5). They interact at the point developed in Section 7.4.

7.2 The Physical Design Basis: The Determinism Posture

The physical design basis is the set of protective functions the venue relies on to keep its crowd safe independently of its networked systems: its structure and its temporary structures; its barriers and pens; its means of egress and the release of their locks and gates; its emergency lighting and wayfinding; its fire detection, suppression and smoke control; its vehicle barriers at the edge; and, as a refuge, its standby power and water. A gate on a means of egress that releases when the fire alarm sounds or its power fails, with no networked controller able to hold it shut, is a candidate for a locally sovereign function. A function meets *Local Sovereignty* §3.2 only if no system an attacker could reach can stop, delay or trigger it, or change its settings except through a governed act.

Whether a function is deterministic decides whether the authorization rules apply to it at all. It does not decide where its actions fall on the Bright Line. Controllers that behave predictably within a stated range of inputs, and that have been formally tested in the deployed configuration, are outside the Command Broker obligation for the actions they take while their inputs stay in that range (*Layer Decomposition and Determinism* §4.3). That is a statement of scope, not of placement; they are outside the obligation only because, and only while, their behavior is bounded by that showing (*The Bright Line* §4.5). The exemption rests on the configuration as installed and tested, and it lapses for a configuration changed outside a governed act

(Determination of 8 October 2026, A Protective Function Running Logic Other Than What Was Tested Is Not Covered by the Carve-Out).

7.3 The AI-Governance Design Basis: Named and Referenced

Machine-learning and generative-AI systems are entering the class: video analytics; crowd-density and anomaly detection; automated access control, face recognition and screening; crowd-flow systems that open and close gates and direct people by signs and displays; automated public address; and building management. The framework names these and places the Bright Line for any action of a component within the scope of *Layer Decomposition and Determinism* §4.3 whose incorrect performance could affect crowd life-safety at or above the severity threshold, naming an automated egress, lockdown or crowd-flow decision, a screening disposition and a life-safety-system actuation, with the architecturally enforced authorization of a qualified security officer or incident commander as Command Broker (framework §5). These systems are governed by the Bright Line and Command Broker architecture (framework §5) and by the Command Broker family, whose parent is CB-Framework_Cross_Sector (*Command Broker Framework*); CB-REG (*Command Broker Family Register*) is authoritative for the family's version state. The coordination class, CorpCo, is the locus at which the sector's placement and obligation are implemented (framework §3.5). Placement is by the consequence of each action, not by the component (*The Bright Line* §4.2, §4.3).

Four cases illustrate the application, and none of them is a placement by component. A system within scope that opens or closes a gate, or directs part of the crowd one way rather than another, takes an action placed on its consequence, which can be to send people toward a crush or toward an attacker; an action whose consequence can reach the threshold requires the authorization of the qualified human the coordination class names as Command Broker, and the release of egress on fire alarm is not taken from the locally sovereign function of Section 7.2 by any crowd-flow system. A system within scope that gives a screening disposition, admitting or turning away a person or a bag, takes with the disposition one action placed on the consequence of admitting it. A system within scope that judges a crowd's density and either raises or withholds an alarm takes an action placed on the consequence of the alarm not raised; a density judgment that suppresses an alarm is placed on that consequence however routine its occasion. And a system within scope that speaks to the crowd, by public address, displays or the event's application, instructing it to move, to stay or that the danger has passed, takes an action placed on the consequence of the crowd moving, or not moving, on it. A disposition within a set of rules already authorized may be discharged in advance under FD-BL-D1 (*Mode of Discharge*); a change to the set itself, being a change to the pre-authorized envelope, is excluded from advance discharge by that determination's rule R3.

Any output on which such an action is grounded carries the indication-integrity obligation of FD-BL-D2 (*Indication Integrity*): a density reading, a video analytic, an identification or a report that the venue is clear, any of which may have been altered or may be wrong, is not a basis for authorization until it can be corroborated. Where it cannot be corroborated through an

independent path, the action it bears on does not proceed without in-the-moment Command Broker authorization, and no placement changes (framework §5). In this class the record of Section 5.2 shows a warning that people were dying reaching the event's structure without producing a stop, which shows what is at stake when an indication and the authority that acts on it are separated.

7.4 The Interaction Point: Crowd Instruction Under Attack

The two design bases meet where the venue's systems act on its crowd while an attacker or a crush is present. The venue's systems must be architecturally incapable of executing an action requiring authorization that originates from a component within the scope of *Layer Decomposition and Determinism* §4.3 without Command Broker authorization. That obligation reaches each such action whose consequence meets the threshold, and the actions most likely to do so are the opening or closing of gates on the crowd's path, an instruction to the crowd to move, stay or disperse, the suppression of a density alarm, and the admission of a person or bag at the screen; each is placed on its own consequence. The physical design basis keeps the crowd able to escape whatever the networks do; the perimeter, the density limit and the stop authority bound the life-safety chain; the AI-governance design basis guarantees that no proposing system can send a crowd toward harm, or silence the warning of it, where the consequence requires it, without a qualified human authorization the architecture can enforce.

7.5 Scope Boundary

This instrument states the physical design basis in full and names the AI-governance design basis without restating it. The Foundational Definitions own the Bright Line, its scope and its obligation; the Command Broker family owns the Command Broker; the coordination class owns the implementation of placement in the sector and the venue's security coordination; Emergency Services owns the response and the mass-care mission; the lodging and retail classes own their functions; and the interdependency analysis owns the convergence consequence. This instrument applies them to the venues at which people gather in their thousands.

8. Cross-Sector Dependencies and Regulatory Context

8.1 What Depends on the Class

The class's outward consequence runs in the three directions the framework names: the lives of the people in the venue, the economic activity that gatherings carry, and the response reserve, the refuge a population needs when a disaster has taken its homes (framework §6.2). The last is felt most when another sector has failed.

8.2 What the Facility Depends On

The facility depends on energy for its lighting, its life-safety systems and, as a refuge, its population's survival; on communications for its instruction of the crowd and its coordination

with the response; on transportation for its crowd's arrival and dispersal, which at a convergence node is the same infrastructure the venue sits upon; and on water for its operation and its refuge (framework §6.1). It depends on Emergency Services for the response to an incident, governed by *Response Forces and Stations* (DBA-ESS-RF-FC2), for answering and dispatch, governed by *Public-Safety Answering and Dispatch* (DBA-ESS-AD-FC1), and for the mass-care mission and the coordination of a wider emergency, governed by *Emergency Management and Coordination* (DBA-ESS-EMCo). And it depends on the healthcare sector for the hospitals that receive its casualties, governed by *Care Delivery* (DBA-HPH-CD-FC1).

8.3 The Boundaries with the Partner Classes and Series

Six boundaries are drawn. With *Lodging and Real Estate* (DBA-CF-LR-FC2) and *Retail and Commerce* (DBA-CF-RT-FC3): hotels, office and residential buildings, shopping centers, gaming and food service are those classes'; where a venue is part of a resort or a mixed-use development, the venue's crowd is this class's and the rest is theirs; the cyberattack on MGM Resorts of September 2023 is held for them. With the coordination class, CorpCo: private venue and event security and its interface with public safety, and the implementation of Bright Line placement in the sector, are its; this class states what the event's plan must contain, and the coordination class supports it. With Emergency Services at the planned gathering: the venue's and the promoter's side is this class's, stated at Section 4.5, and the response's side is that series' (*Response Forces and Stations* §4.7, §8.3). With Emergency Services in the disaster-support role: the venue and its capacity to host are this class's, and the mass-care mission is that series' and Emergency Support Function 6's (framework §6.3). With every co-located sector at a convergence node: the venue's crowd and its egress are this class's, and the aggregate consequence is the interdependency analysis' (framework §6.3). With the healthcare sector: the casualties' distribution to hospitals is a shared interface between the response and the hospitals, and this class's part is that its medical provision brings casualties to the point where the response can take them.

8.4 Regulatory and Institutional Context

The class is privately owned, open to the public by design and governed primarily by local building and fire codes rather than by a federal operating regulator (framework §2.1). The Department of Homeland Security is the Sector Risk Management Agency, the role the framework calls Sector-Specific Agency, and CISA's work on securing soft targets and crowded places is the nearest federal security framework. The life-safety codes the jurisdictions adopt govern assembly occupancies, their occupant loads and their egress, and the National Fire Protection Association's *Life Safety Code* requires trained crowd managers at assembly occupancies above stated sizes. Events on open ground are governed by the permits the jurisdictions issue, which the Texas task force found inconsistent across that state. The Support Anti-terrorism by Fostering Effective Technologies Act of 2002 offers liability protection for qualified anti-terrorism technologies and services, which some venues' security programs have

obtained. The mass-care mission is organized under Emergency Support Function 6 of the National Response Framework. Uncrewed aircraft over gatherings are governed by the Federal Aviation Administration's airspace rules, and the authority to act against them is held by a small number of federal agencies and by state and local law enforcement agencies certified under the SAFER SKIES Act of 2025.

Abroad, the United Kingdom's Terrorism (Protection of Premises) Act 2025, which received Royal Assent on 3 April 2025, places a duty on those responsible for qualifying premises and events to prepare for and protect against terrorist attack, under the Security Industry Authority as regulator; it descends from the recommendation of the Manchester Arena inquiry (Section 5.3), and it is cited here as a statement of what a jurisdiction has required, not as a standard for this class.

The regulatory gap is the framework's (framework §2.3). The life-safety codes are calibrated to accidental events, and they measure occupancy and egress at the doors. No instrument in the United States requires a venue to set its perimeter by its crowd rather than its building, to state who acts in a shared space, to state the density limit of a crowd and how it is held, or to place the authority to stop a show with a person who can see the crowd; and no instrument requires a venue designated as a refuge to show that it can withstand the hazard it shelters people from and sustain them independent of the systems that hazard takes. This class states those as design-basis requirements for the tiers and overlays that carry the load.

9. Inheritance Map and Anticipated Derivatives

9.1 The Inheritance Map

The series requires every facility-class instrument to state, at first publication, which elements it inherits unchanged, which it inherits with adaptation, and which it defines locally. Facility-class instruments add the facility-function-specific consequence chain and, where applicable, the disaster-support and convergence-node consequences (framework §8.1). Table 9-1 is the map for this class. Where the framework is a skeleton, the element is developed here and recorded for the framework to take up.

Table 9-1. Inheritance map for DBA-CF-PA-FC1.

Element	Source in the framework	Inheritance mode
Function-primary placement, PA function class	§1.3, §3.1, §3.2	Inherited unchanged
Occupancy and convergence sub-index; disaster-support and convergence overlays	§3.1, §8.2	Inherited with adaptation; three occupancy tiers and two separate overlays (Section 2.2)

Element	Source in the framework	Inheritance mode
Design-basis principles (consequence severity, not likelihood)	§4.1	Inherited unchanged
Three consequence treatments	§1.5, §3.6	Inherited; stated as Chains L, R and N (Section 3.2)
Crowd density, perimeter reach and stop time	§2.3, §3.6	Locally defined as parameters (Section 3.3); for the framework to take up
Event categories	§4.1 (candidates)	Developed here ahead of the framework; for the framework to take up
Bright Line and Command Broker architecture	§5; FD-BL; Command Broker family	Inherited with adaptation: gates, screening dispositions, density alarms and crowd instructions placed on their consequence (Section 7.3)
Planned-gathering interface, the venue's side	§6.3, §9	Locally stated in full, as the framework owed (Section 4.5)
The crowd, the perimeter and the refuge	§3.2, §1.5	Locally defined (Section 4)
Safe Assembly and Sustained Refuge Conditions	§7 (skeleton)	Locally defined (Section 6.7)
Declared residuals and conditionality check	not yet in the framework	Locally defined; recorded for the framework (Sections 6.8, 6.9)
Boundaries with Emergency Services and the interdependency analysis	§6.3	Inherited unchanged and applied (Section 8.3)

9.2 Anticipated Derivatives

Venue- and event-specific design bases inherit this class document by reference and add only what is locally specific, each carrying a recorded applicability determination for any scope it reduces. The three reference events this class proposed at v0.1, on the author's decision of 14 October 2026, are conferred by the framework at §9.1: DBA-CF-RE-1, *The Bombing at the Manchester Arena*, 22 May 2017 (Section 5.3); DBA-CF-RE-2, *The Crowd Crush at the Astroworld Festival*, Houston, 5 November 2021 (Section 5.2); and DBA-CF-RE-3, *The Louisiana Superdome as Refuge of Last Resort in Hurricane Katrina*, August and September 2005 (Section 5.5). Their instruments were written by Project 11 on 14 October 2026, and this

revision takes in their erratum and six recommendations. The plan for closing the seven sectors named the cyberattack on MGM Resorts in September 2023 for this sector; on the author's decision of 14 October 2026 it is held for *Lodging and Real Estate* (DBA-CF-LR-FC2) or *Retail and Commerce* (DBA-CF-RT-FC3), whose register entries hold hotels and gaming. The destruction of the drama theater in Mariupol on 16 March 2022 is noted and not proposed (Section 6.3); the shooting at the Route 91 Harvest festival is already the Emergency Services series' reference event (Section 5.4). The facility-class use case for the governing military-action scenario of Section 6.3 is DBA-CF-UC-1, issued under this revision in the fifth wave of the plan.

10. Gaps and Open Items

10.1 Quantitative Design Parameters

The six parameters of Section 3.3 are stated without values. Their values for a venue or event are to be validated against its occupancy and egress calculations, its event plan and permit, the assessments CISA offers to venue owners and, for a refuge, the jurisdiction's mass-care plan. This instrument is written to be published, and states none; a venue's design basis that states its perimeter arrangements and its weaknesses should be held with the care that information requires.

10.2 The Convergence Consequence

The framework declines to measure the convergence consequence at the scale of a single facility and hands it to the cross-sector interdependency analysis (framework §3.6, §6.3). This instrument states its chain and its parameter, the convergence count, and declares it as a residual (Sections 3.2, 3.3, 6.8). Whether the interdependency analysis can take the parameter as its entry point is for that analysis, and the corpus maintains it separately.

10.3 The Crowd as Hazard

The framework's life-safety measure is occupancy and egress under a deliberate-event condition (framework §3.6). This instrument adds crowd density, and the authority to relieve it, because the record of Section 5.2 shows a crowd that killed with no attacker and no failure of egress at the doors. Whether the framework widens its measure is for the framework.

10.4 Local Sovereignty in the DBA-MA Sense

Section 6.7 relies on locally sovereign protective functions for egress and fire protection, as FD-LS defines the property. Whether the DBA-MA parent framework's local-sovereignty principle is the same thing FD-LS defines remains the open question the other classes record.

10.5 The Agency's Name

The framework names the Department of Homeland Security the Sector-Specific Agency (framework §1.4, §2.1). The role has since been renamed Sector Risk Management Agency, and

this instrument uses the current name. The framework's usage is the framework's to conform at its next revision.

10.6 The Reference Events

The three reference events this class proposed are conferred by the framework at §9.1 as DBA-CF-RE-1 to DBA-CF-RE-3, and their instruments were written by Project 11 on 14 October 2026. Manchester is recorded as a non-military adversarial event, a single act, with Transportation an impacted sector; Astroworld as non-adversarial, the grand jury's finding that no crime occurred followed under RECORD PRIMACY, with Emergency Services an impacted sector; and the Superdome as non-adversarial, with Emergency Services and Transportation impacted sectors. The instruments name neither the attacker nor those who died, refer to officials and staff by role, and state each source's figures as its own. This revision takes in the erratum at Section 5.5 and the recommendations at Sections 3.3, 4.4, 4.5 and 4.6. The use case for the governing scenario of Section 6.3 is DBA-CF-UC-1.

10.7 The Register Row

The DBA Facility Class Register now records this class at v0.1 as an issued working draft under the framework at v0.4; Project 0 entered the row on 14 October 2026.

10.8 The Framework's Section 9

The framework's Section 9 records that the full statement of the venue's side of the planned-gathering interface is owed by this class (framework §9). Section 4.5 states it. The framework's record is the framework's to close at its next revision, which the conferral will be.

10.9 What This Class Notes to Other Instruments

Several elements of this instrument belong to other instruments, and this instrument records them as noted without stating them for those instruments. To the framework: the three chains (Section 3.2), the six parameters and crowd density in particular (Sections 3.3, 10.3), the occupancy tiers and the two overlays stated separately (Section 2.2), the event categories of Section 5, the two adequacy conditions of Section 6.7, the declared residuals of Section 6.8, the conditionality principle of Section 6.9, the agency's name (Section 10.5), and the closing of its Section 9 record (Section 10.8); the conferral section it also asked for is now framework §9.1. To Emergency Services: that the venue's side of the planned-gathering interface is now stated (Section 4.5), and that the Manchester inquiry's second volume is a source for the response's side. To *Lodging and Real Estate* and *Retail and Commerce*, when written: the boundaries of Section 8.3 and the MGM event held for them. To the interdependency analysis: the convergence count (Section 10.2). From the instruments: to the coordination class, CorpCo, when written, *The Bombing at the Manchester Arena* R-1 and R-2 on the watch at the edge and the report time, and *The Crowd Crush at the Astroworld Festival* R-1 on the triggers that stop the show; to *Response Forces and Stations*, *The Crowd Crush at the Astroworld Festival* R-1 and R-2; to *Emergency Management*

and Coordination, *The Crowd Crush at the Astroworld Festival* R-2 and *The Louisiana Superdome as Refuge of Last Resort in Hurricane Katrina* R-1 and R-2; and to the Transportation series, *The Louisiana Superdome as Refuge of Last Resort in Hurricane Katrina* R-2 on the means of evacuation. The register row is entered (Section 10.7).

References

DBA-CF Series and Parent Framework

Commercial Facilities Sector Framework (DBA-CF_Sector_Framework). Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued.

DBA-MA Parent Framework. Eclectic Technologies. Source of the design-basis-under-military-action methodology. FD-DBL (*Design Basis Layers*), held for the edition following the October 2026 Edition, is the canonical statement of the four military-action characteristics (§4.4, §6.4).

Defense in Depth (DBA-MA-DiD). Eclectic Technologies, 2026. Declared-residual output form (§3.5).

Government Operations and Continuity (DBA-GF-GO-FC1), *Care Delivery* (DBA-HPH-CD-FC1), *Response Forces and Stations* (DBA-ESS-RF-FC2), *Public-Safety Answering and Dispatch* (DBA-ESS-AD-FC1) and *Emergency Management and Coordination* (DBA-ESS-EMCo). Eclectic Technologies, 2026. The structural models, the response's side of the planned gathering, dispatch, and the mass-care mission. *The Route 91 Harvest Shooting of October 2017* (DBA-ESS-RE-11), held by Project 11.

Facility-Class Types Are Physical, Coordination and Logical, author's determination of 30 September 2026, and *A Protective Function Running Logic Other Than What Was Tested Is Not Covered by the Carve-Out*, author's determination of 8 October 2026, both recorded by Project 0. *Closing the Seven Sectors*, working plan of 13 October 2026, Project 9.

The Bombing at the Manchester Arena (DBA-CF-RE-1), *The Crowd Crush at the Astroworld Festival* (DBA-CF-RE-2) and *The Louisiana Superdome as Refuge of Last Resort in Hurricane Katrina* (DBA-CF-RE-3). Eclectic Technologies, 2026. Held by Project 11; conferred at Commercial Facilities Sector Framework §9.1 and cited by short title and finding or recommendation number. The RECORD PRIMACY convention of the DBA Reference Event Register, Project 0.

Foundational Definitions and Command Broker Family

FD-BL, FD-BR, FD-LD, FD-LS, FD-EV, FD-BL-D1 and FD-BL-D2, cited by designation; version state in FD-REG (*Governing Stack Register*). CB-Framework_Cross_Sector (*Command Broker Framework*), cited by designation; version state in CB-REG (*Command Broker Family Register*).

Regulatory, Standards and Guidance

National Fire Protection Association. NFPA 101, *Life Safety Code*, assembly occupancies. Cybersecurity and Infrastructure Security Agency. *Commercial Facilities Sector* and *Securing Soft Targets and Crowded Places*. Support Anti-terrorism by Fostering Effective Technologies Act of 2002. SAFER SKIES Act, National Defense Authorization Act for Fiscal Year 2026, December 2025. Federal Emergency Management Agency. *National Response Framework*, Emergency Support Function 6, Mass Care. United Kingdom. Terrorism (Protection of Premises) Act 2025.

Events and Investigations

Manchester Arena Inquiry. *Report*, Volume 1, *Security for the Arena*, June 2021; Volume 2, *Emergency Response*, November 2022; Volume 3, *Radicalisation and Preventability*, March 2023.

Texas Task Force on Concert Safety. *Report*, April 2022. Houston Police Department, report of its investigation of the Astroworld festival, July 2023, as reported. Harris County District Attorney, statement of the grand jury's finding, June 2023. Harris County Institute of Forensic Sciences, findings on cause of death, December 2021, as reported.

United States House of Representatives, Select Bipartisan Committee to Investigate the Preparation for and Response to Hurricane Katrina. *A Failure of Initiative*, February 2006. United States Senate, Committee on Homeland Security and Governmental Affairs. *Hurricane Katrina: A Nation Still Unprepared*, May 2006.

Amnesty International, investigation of the strike on the Donetsk Regional Drama Theater in Mariupol, 30 June 2022; Associated Press, investigation of the same, May 2022; report under the Organization for Security and Co-operation in Europe, April 2022; statements of the Mariupol City Council and of the Russian Ministry of Defense, March 2022, as reported.

Revision History

Version	Date	Author	Description
0.1 — DRAFT	October 2026	Tim Roxey	Initial issue. First facility-class instrument in the DBA-CF series, typed physical under the author's determination of 30 September 2026, in the ten-section structure of DBA-HPH-CD-FC1 and DBA-GF-GO-FC1. Written under the plan Closing the Seven Sectors, wave 4, on the author's decisions of 14 October 2026: Public Assembly and Mass-Gathering Venues as the sector's first class; three reference events proposed, the bombing at the Manchester Arena (2017), the crowd crush at the Astroworld festival (2021) and the Louisiana Superdome as refuge of last resort during Hurricane Katrina (2005), the last in place of the cyberattack on MGM Resorts (2023), which is held for the lodging

Version	Date	Author	Description
			and retail classes. States the framework's three consequence treatments as Chains L, R and N, adds crowd density, perimeter reach and stop time as parameters, states the venue's side of the planned-gathering interface in full as the framework owed, and states two adequacy conditions, Safe Assembly and Sustained Refuge, with the convergence consequence declared and handed to the interdependency analysis. Checked before issue by an independent review agent, not a person, against primary sources and for internal consistency; its substantive findings were taken in, among them the Texas task force's occupancy finding corrected to its own words (no occupancy load had been issued), the extension of counter-drone authority to certified state and local law enforcement recorded, crowd instructions placed on their consequence rather than on their kind, and an unsupported statement about the Astroworld record removed.
0.2 — DRAFT	October 2026	Tim Roxey	Conforms to the framework at v0.5 and takes in the return of Project 11's Commercial Facilities batch. The three reference events proposed at v0.1 are stated as conferred, as DBA-CF-RE-1 (Manchester), DBA-CF-RE-2 (Astroworld) and DBA-CF-RE-3 (the Superdome), at Sections 1.3, 5.2, 5.3, 5.5, 9.2 and 10.6, Astroworld under RECORD PRIMACY. Erratum 1 attributes the Senate committee's statement of the Saturday to New Orleans as a whole and records the Superdome's last departures on Saturday 3 September (Section 5.5). The optional refinements add the reconnaissance on at least three occasions and the officers told to be in the City Room, the task force's placing of stop authority and the times of the evening, the county and city split, each source's figure for the number sheltered and the Monday assurance of five hundred buses. The six recommendations enter at Sections 3.3 (the watch confirmed; a report time; the admission stop and the overflow; evacuation means committed before opening), 4.4 (triggers that stop the show), 4.5 (the plan binding every authority), 4.6 and 6.3. Section 10.9 carries the notings to CorpCo, Response Forces and Stations, Emergency Management and Coordination and the Transportation series. The cover lineage stays as issued. No change to the class's scope, chains, parameters, event categories or residuals; the v0.1 entry is left as a record.